

Modern Type Theory

An introduction to modern dependent type theory

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Π -types

The **dependent function type**, also known as *dependent products* or **Π -types**, contain functions whose codomain type may vary based on the terms of its domain.

Π -types are formed as follows:

$$\frac{\Gamma \vdash A \text{ type} \quad \Gamma.A \vdash B \text{ type}}{\Gamma \vdash \Pi(A, B) \text{ type}}$$

$$\frac{\Gamma \vdash A \text{ type} \quad \Gamma.A \vdash b : B}{\Gamma \vdash \lambda(b) : \Pi(A, B)}$$

$$\frac{\Gamma \vdash a : A \quad \Gamma.A \vdash B \text{ type} \quad \Gamma \vdash f : \Pi(A, B)}{\Gamma \vdash f \ a : B[\text{id}.a]}$$